

LABORATORY 1

CMOS INVERTER ANALYSIS

OBJECTIVES

No.	Objectives	Requirements
1	Knowing the trip-point of an inverter and the concept of PVT	Find the trip-point of the inverter and explain the variation with PVT.
2	Understand the calculator functions working	Using calculator functions to measure some parameters of the inverter with transient analysis.
3	Understand the fanout effect	Calculate the transient performance of inverter with multiple fanout condition
4	Design a high speed 4-to-1 multiplexer	Design a transistor-level 4-to-1 multiplexer using CMOS logic for high-frequency operation, @5GHz, and meet the signal integrity specification.

PREPARATION FOR LAB 1

- Review the knowledge about an CMOS inverter.
- Review the knowledge about using Virtuoso and ADE-Explorer to simulate.

LAB 1 INFORMATION

- Consider *NMOS nmos1v_lvt device*, *PMOS pmos1v_lvt device* in *gpd45nm* and power supply $vdd = 1.0V$
- **Note:** All figures in the document are for reference only.



EXPERIMENT 1

1.1. Objective: Build a testbench to measure the trip-point of the inverter. Setup and simulate design with ADE-Explorer environment.

1.2. Requirements: Simulate with dc analysis with ADE-Explorer environment.

1.3. Instruction: Assemble the circuit as shown in **Figure 1** and **Figure 2**, setting parameters for power supplies (using **vdc** source), with $L_n = L_p = 45nm$, $C_{load} = 100fF$, $v_{dd} = 1.0V$

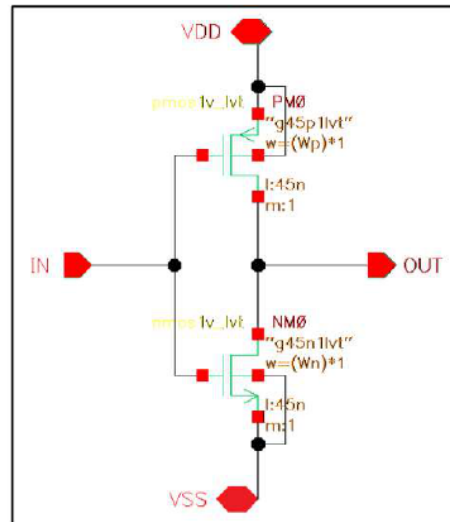


Figure 1. Schematic of INV

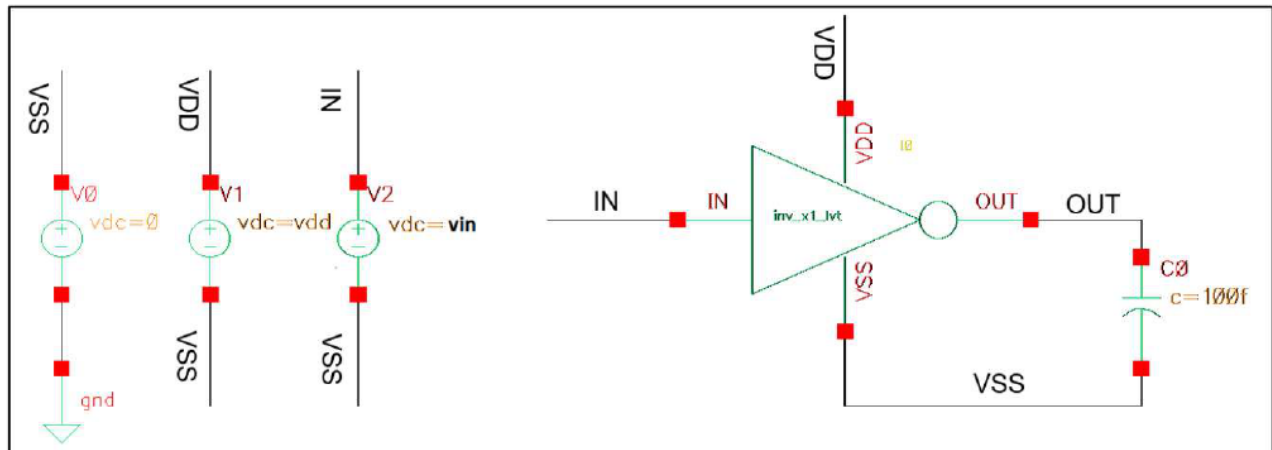


Figure 2. Testbench dc analysis of INV

- Created testbench and ADE-Explorer similar to the instruction in LAB0.
- At tab **Analyses** → Choose the type of the analysis (dc, ac, tran, noise, stb, pss ...) as **Figure 3**. In this experiment, we use **DC analysis** to find the trip-point of INV from in-out characteristic curve.

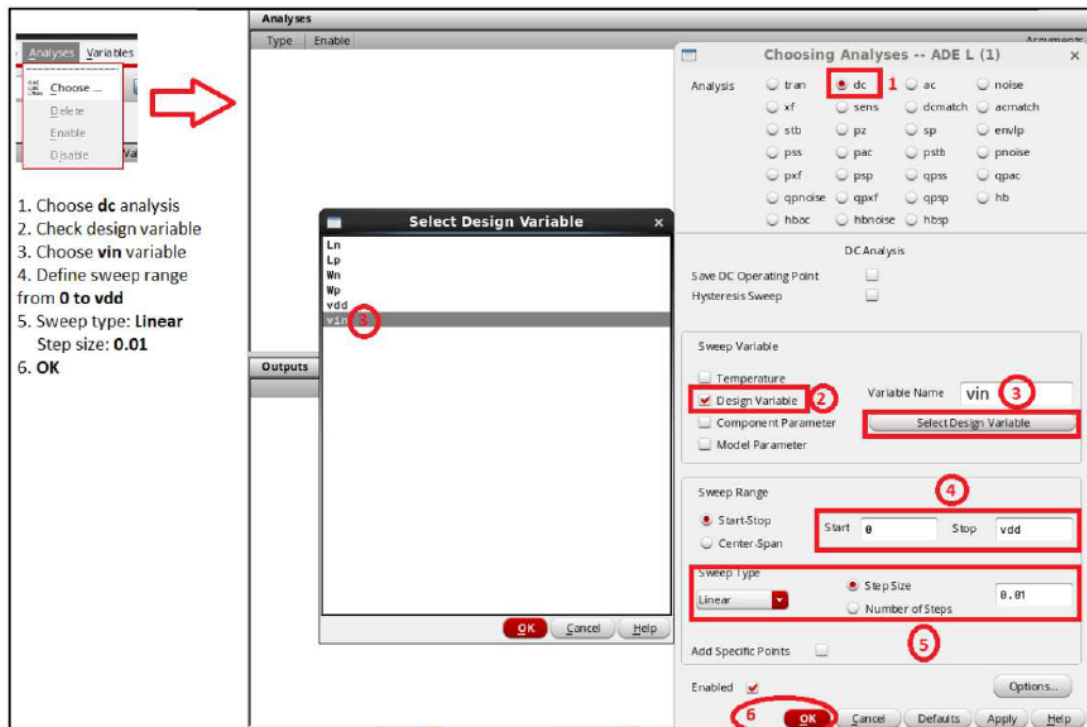


Figure 3. Setup the analysis method

- We can re-define value of variable at **Global Variables** tab as **Figure 4**.

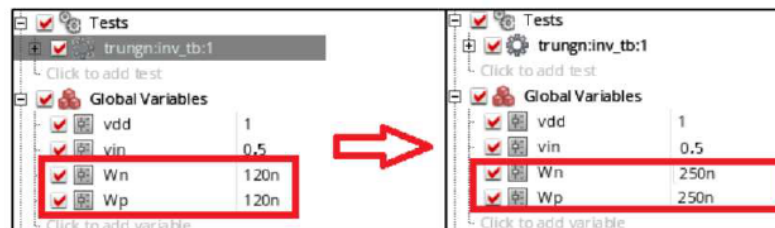


Figure 4. Change value of variables

- Multiple variables can also be swept in ADE Explorer. As shown in **Figure 5**, there are two ways to define this, both producing the same results for W at four values: 250nm, 500nm, 650nm, and 1000nm.
 - Option 1: List value: **val1,val2,val3,val4,val5,...,valn**
 - Option 2: Define the list attribute: **start : step : stop**

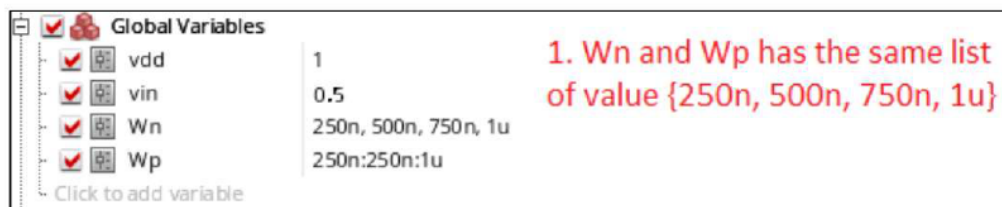


Figure 5. Multiple variables sweep setup

- After finished setup the testbench. At **Run** tab → Choose **“Single Run, Sweeps and Corner”** to run the testbench.
- At **Results** tab → Choose **/IN** and **/OUT** signal → Right click → **Plot/Plot All** as **Figure 6**.

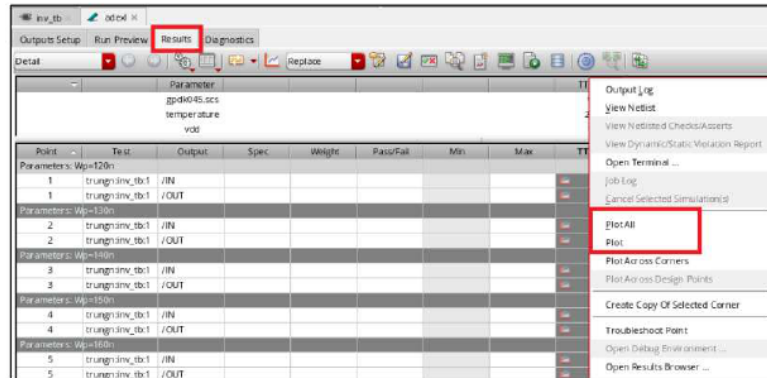


Figure 6. Plot waveform

1.4. Analyze and Report:

- Trip point defines the input voltage that the output changes the state (from 0 to 1 / from 1 to 0).
- In switching state working, the inverter needs to balance between the rising and falling edge of the inverter to ensure stability. So, we need to **find the P/N ratio size of PMOS and NMOS (W_N and W_P) to get the balanced state (or balance trip point) at TTNN corner.**
- Verify the size of NMOS and PMOS (W_N and W_P) to get the balance trip-point with minimum size.
- **Writing a calculator function to measure the trip-point.**

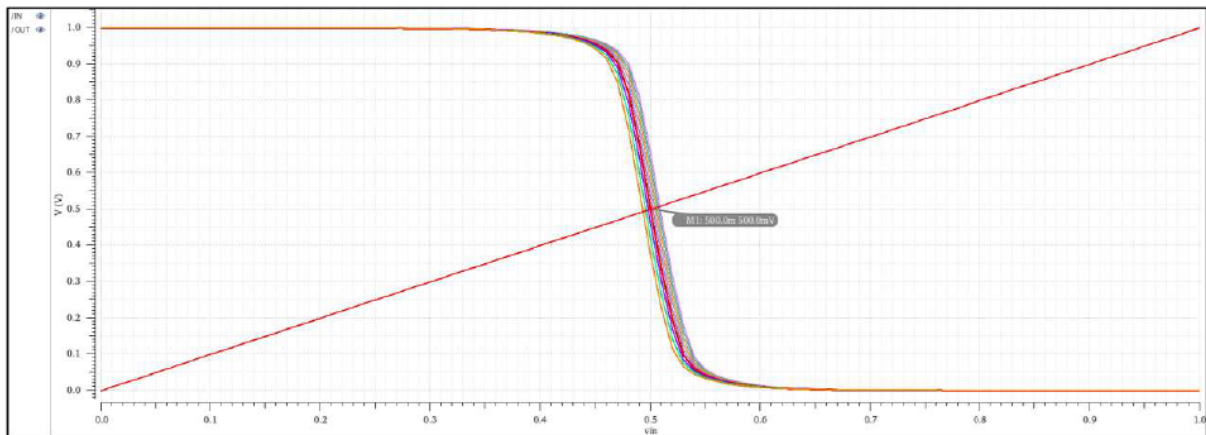
➤ 1.4.1. Testcase:

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT
Temp	25°C
V_{DD}	1 V
L_n	45nm
W_n	

<i>Lp</i>	45nm
<i>Wp</i>	
<i>Observe</i>	Inverter IN-OUT characteristic curve
<i>Trip-point Equation</i>	
<i>Trip-point value</i>	

- Please provide the schematic and testbench here.

➤ 1.4.2. Waveform:



Parameter							TTNN
gpdK045.scs							tt
temperature							25
vdd							1
Test	Output	Spec	Weight	Pass/Fail	Min	Max	TTNN
trungn:inv_tb:1	v_tp				405.1m	546m	500.1m

Figure 7. Example waveform and calculator result finding the balance trip-point

➤ 1.4.3. Observe from the waveform:

Wn	Wp	Trip-point Value
120n	120n	
120n	140n	
120n	160n	
...	...	
120n	300n	

140n	120n	
140n	140n	
140n	160n	
...	...	
140n	300n	
...	...	

➤ 1.4.4. Explanation (if applicable):

■

✚ Using the W_N and W_P for **Experiment 2** and **Experiment 3**.



EXPERIMENT 2

2.1. Objective: Consider the variation of trip point with the changing of PVT

2.2. Requirements: Simulate dc analysis with ADE-Explorer environment.

2.3. Instruction: Assemble the circuit as shown in **Figure 2**, setting parameters for power supplies (using **vdc** source), with $L_n = L_p = 45nm$, $C_{load} = 100fF$, $vdd = 1.0V$, W_N, W_P in **Experiment 1**.

- You can create some other corners to consideration as **Figure 8**.

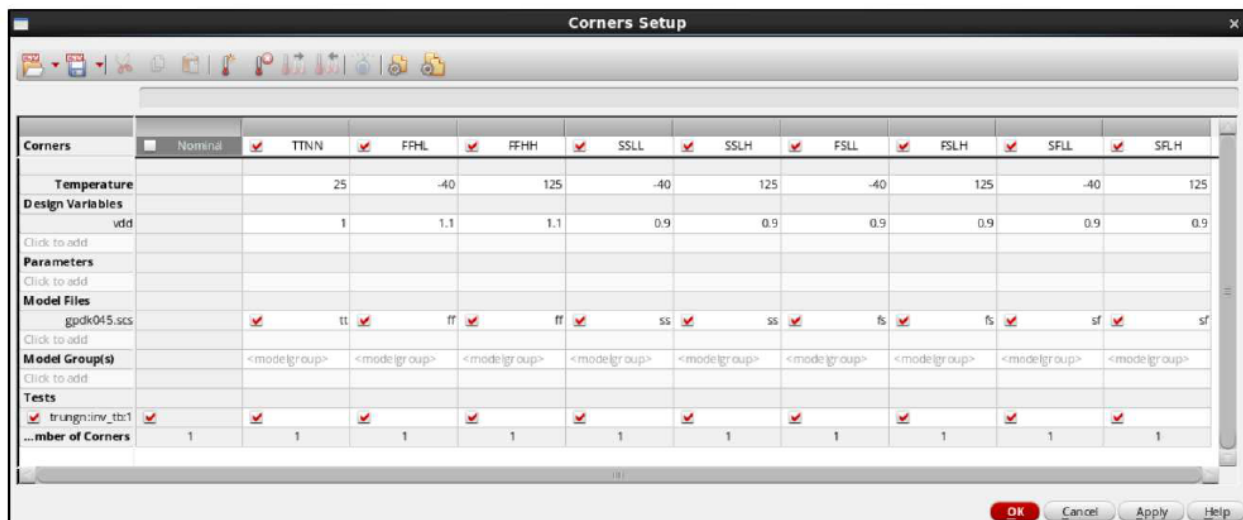


Figure 8. Setup corner

- With TTNN corner, the first symbol **T (Typical)** represents for NMOS device, the second **T (Typical)** symbol represents for PMOS device, the third symbol **N (Nominal)** represents for supply voltage and the last symbol **N (nominal)** represents for temperature.
- With NMOS/PMOS device, we have total 5 common representatives **TT, FF, SS, SF, FS** (**F – Fast, S – Slow**).
- With supply voltage, we use symbol **N (Nominal)**, **L (Low)** and **H (High)** to mark.
- With temperature, we use symbol **N (Nominal)**, **L (Low)** and **H (High)** to mark.
- Click **OK** to save Corner setup.
- In **LAB0**, the rising time of the inverter is defined using an equation

$$\text{riseTime}(\text{VT}("/\text{OUT}") \text{ } 0 \text{ nil } 1 \text{ nil } 10 \text{ } 90 \text{ nil "time" }) \quad (1)$$
- But equation (1), we **fixed the value of vdd = 1.0V** → Cannot use this equation for all corner when measure rising time.

- We have to define the variable **vdd** in the measurement equation to measure the rising time at many corners by using syntax **VAR("vdd")** as Equation (2)

`riseTime(VT("/OUT") 0 nil VAR("vdd") nil 10 90 nil "time")` (2)

2.4. Analyze and Report:

➤ 2.4.1. Testcase:

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT, FF, SS, SF, FS
Temp	-40°C, 25°C, 125°C
V_{DD}	0.9V, 1 V, 1.1V
Ln	45nm
Wn	 (Value in Experiment 1)
Lp	45nm
Wp	 (Value in Experiment 1)
Observe	Inverter IN-OUT characteristic curve
Trip-point Equation	

- Please provide the schematic and testbench here (Capture setup corners as Figure 8).
- Setup testbench with 9 PVT corners in **Table 1** with ADE-Explorer.

Corner	TTNN	FFHL	FFHH	SSLL	SSLH	FSLH	SFLH	SFLL	SFLH
<i>vdd</i>	1.0	1.1	1.1	0.9	0.9	0.9	0.9	0.9	0.9
<i>Temp</i>	25	-40	125	-40	125	-40	125	-40	125

Table 1. PVT corner definition

➤ 2.4.2. Waveform:

Detail															
Parameter	TTNN	FFHL	FFHH	SSL	SSLH	FSLL	FSLH	SFLL	SFLH						
gpdK45.scs	ff	ff	ss	ss	ss	fs	fs	sf	sf						
temperature	25	-40	125	-40	125	-40	125	-40	125						
vdd	1	1.1	1.1	900m	900m	900m	900m	900m	900m						
Test	Output	Spec	Weight	Pass/Fail	Min	Max	TTNN	FFHL	FFHH	SSL	SSLH	FSLL	FSLH	SFLL	SFLH
trungninv.tb1	v_tp				405.1m	546m	500.1m	544.1m	546m	448.6m	463.7m	405.1m	410.9m	491.2m	500.1m

➤ 2.4.3. Observe from the waveform:

➤ 2.4.4. Explanation (if applicable):

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EXPERIMENT 3

3.1. Objective: Using the Calculator function to measure the performance of inverter.

3.2. Requirements: Simulate tran analysis with ADE-Explorer environment.

3.3. Instruction:

- To copy the ADE-Explorer setup and re-define “Analysis method”, “Variables” or “Output signal/Measurement” based on your recent setting. You need to choose “**Open Existing View**” instead of “**Create New View**” as **Figure 11**.

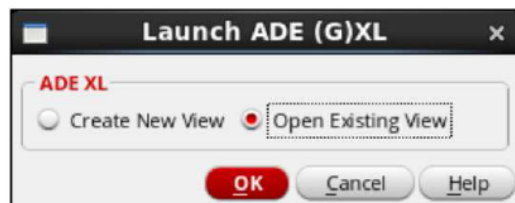


Figure 11. Option with copied adexl setup

- Assemble the circuit as shown in **Figure 12**, setting parameters for power supplies (using **vpulse** source), with $L_n = L_p = 45\text{nm}$, $C_{load} = 100\text{fF}$, $vdd = 1.0\text{V}$ W_N , W_P in **Experiment 1**.
- Set the variable for vpulse source parameter (voltage 1, voltage 2, period, rise time, fall time, pulse width) in **Figure 13**.

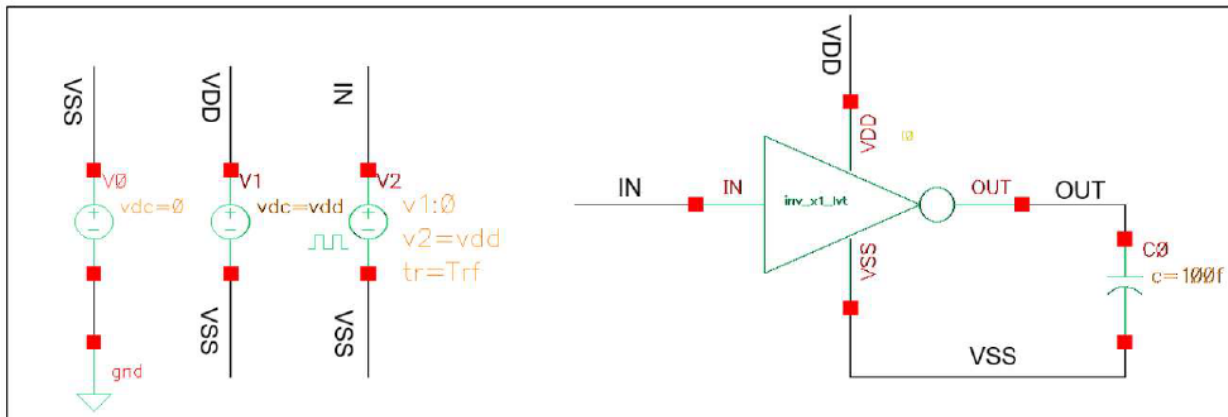


Figure 12. Testbench tran analysis of INV

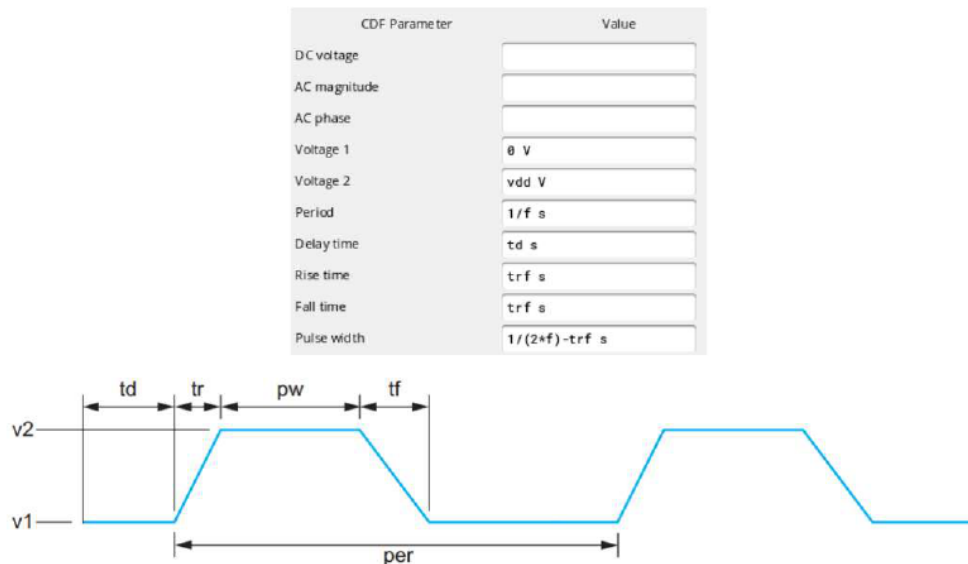


Figure 13. Testbench tran analysis of INV

- Define the value of variable in the ADE-Explorer variable setting

$$F = 5\text{GHz}, Td = 10\text{ps}, Trf = \frac{1}{10 * F}, Pulse_width = \frac{1}{2 * F} - Trf$$

- Set up the transient analysis as LAB0 (consider the tstop time for transient simulation).

3.4. Analyze and report:

- 3.4.1. Testcase with Cload = 100fF:

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT, FF, SS, SF, FS
Temp	-40°C, 25°C, 125°C
V _{DD}	0.9V, 1 V, 1.1V
Ln	45nm
Wn	 (Value in Experiment 1)
Lp	45nm
Wp	 (Value in Experiment 1)
Frequency	5GHz
Slew time	10%*T _{cycle}
C _{load}	100fF

- Setup testbench with 9 PVT corners in **Table 1** with ADE-Explorer.

- Please provide the schematic and testbench here.

➤ 3.4.2. Waveform and Explanation with $C_{load} = 100fF$:

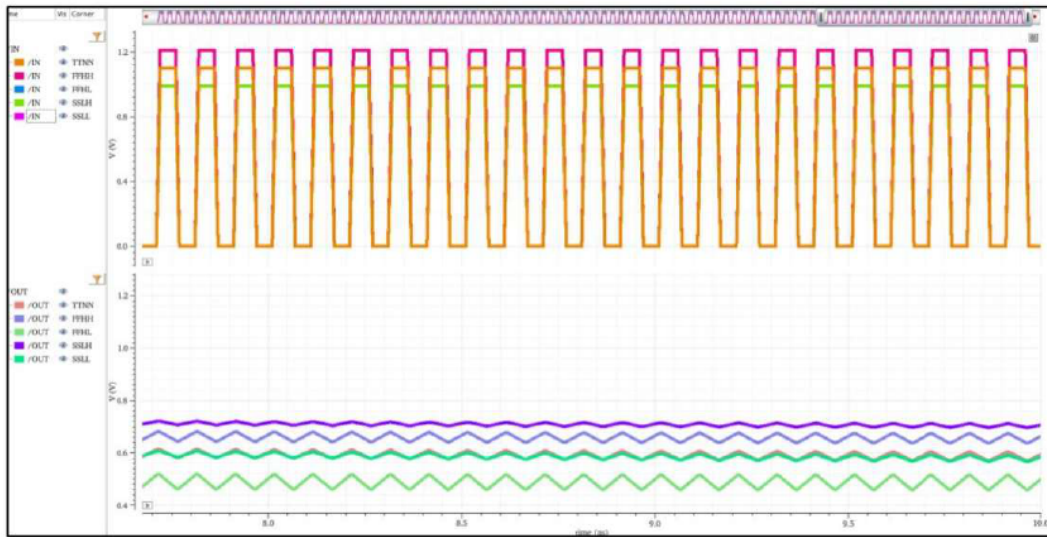


Figure 14. Result waveform of inverter simulation with transient analysis with $C_{load} = 100fF$

- Explain why the inverter cannot run at 5 GHz in the testbench.

➤ 3.4.3. Testcase with $C_{load} = 1fF$:

- Change C_{load} from 100fF to 1fF and rerun the simulation. Setup testbench with 9 PVT corners in **Table 1** with ADE-Explorer.

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT, FF, SS, SF, FS
Temp	-40°C, 25°C, 125°C
V_{DD}	0.9V, 1V, 1.1V
L_n	45nm
W_n	 (Value in Experiment 1)
L_p	45nm
W_p	 (Value in Experiment 1)
Frequency	5GHz
Slew time	10%* T_{cycle}
C_{load}	1fF

➤ **3.4.4. Waveform and Explanation with Load = 1fF:**

- With Load = 1 fF, the waveforms exhibit correct behavior under all PVT conditions

as Figure 15

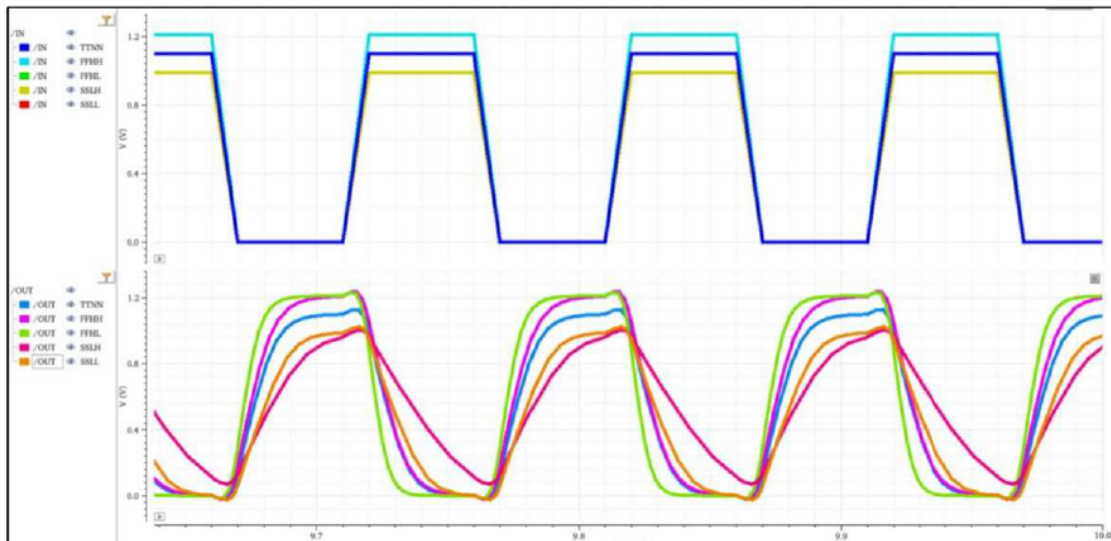


Figure 15. Result waveform of inverter simulation with transient analysis with Load = 1fF

- Create the calculator function to measure rising time, falling time, duty cycle, delay from input to output, and current of inverter in **Table 2** with the guideline in **Figure 16** (example only).

Note: You need write *all measurements* listed in **Table 2**, do not just refer from **Figure 16** and **Figure 17**.

T_rise_avg	expr	riseTime(VI("/OUT") 0 nil VAR("vdd") nil 10 90 nil "time")
T_rise	expr	delay(7wf1 VI("/OUT") ?value 1 (0.1 * VAR("vdd")) ?edge1 "rising" ?nth1 10 ?td1 0.0 ?wf2 VI("/OUT") ?value 2 (0.9 * VAR("vdd")) ?edge2 "rising" ?nth2 10 ?td2 nil ?stop nil ?multiple nil)
T_fall_avg	expr	fallTime(VI("/OUT") VAR("vdd") nil 0 nil 90 10 nil "time")
T_fall	expr	delay(7wf1 VI("/OUT") ?value 1 (0.9 * VAR("vdd")) ?edge1 "falling" ?nth1 10 ?td1 0.0 ?wf2 VI("/OUT") ?value 2 (0.1 * VAR("vdd")) ?edge2 "falling" ?nth2 10 ?td2 nil ?stop nil ?multiple nil)

Figure 16. Guideline to configure the parameters for measurement functions

- Setup testbench with 9 PVT corners in **Table 1**.

Parameter							TTNN	FFHL	FFHH	SSLL	SSLH	FSLL	FSLH	SFLL	SFLL
gpd4045.scs							tt	ff	ff	ss	ss	fs	fs	sf	sf
temperature							25	-40	125	-40	125	-40	125	-40	125
vdd							1	1.1	1.1	900m	900m	900m	900m	900m	900m
Test	Output	Spec	Weight	Pass/Fail	Mn	Max	TTNN	FFHL	FFHH	SSLL	SSLH	FSLL	FSLH	SFLL	SFLL
trungninv.tb:2	/IN														
trungninv.tb:2	/OUT														
trungninv.tb:2	/11/VDD														
trungninv.tb:2	T_rise_avg				24.23p	67.6p	35.69p	24.23p	34.15p	47.68p	67.6p	40.83p	61.13p	31.42p	46.08p
trungninv.tb:2	T_rise				23.99p	67.61p	35.68p	23.99p	34.16p	48.32p	67.61p	41.15p	60.93p	31.42p	46.05p
trungninv.tb:2	T_fall_avg				24.23p	67.6p	35.69p	24.23p	34.15p	47.68p	67.6p	40.83p	61.13p	31.42p	46.08p
trungninv.tb:2	T_fall				17.04p	69.8p	28.51p	17.04p	28.2p	38.82p	69.8p	23.28p	40.53p	32.19p	58.40p

Figure 17. Example result of measurement parameter with transient analysis

 Annotate some measurement parameters

Item	Detail
Trise_avg	Rising time from low voltage to high voltage (10%-90%) measured by risetime function
Tfall_avg	Falling time from high voltage to low voltage (90%-10%) measured by falltime function
DT_H	Duty cycle of high voltage measuring by delay function
DT_L	Duty cycle of low voltage measuring by delay function
Td_R2F	Delay from rising input to falling output measured by delay function
Td_F2R	Delay from rising input to falling output measured by delay function
Trise	Rising time from low voltage to high voltage (10%-90%) measured by delay function
Tfall	Falling time from high voltage to low voltage (90%-10%) measured by delay function
I_static_0	Static current with low voltage input
I_static_1	Static current with high voltage input
I_dynamic	The average dynamic current of the inverter

Table 2. Parameter of inverter performance explanation

- **Complete the table with the results from Calculator:**

Corner	Trise_avg (ps)	Tfall_avg (ps)	Trise (ps)	Tfall (ps)	DT_H (%)	Td_R2F (ps)	Td_F2R (ps)	I_static_ _0	I_static_ 1	I_dyna mic
TTNN										
FFHH										
FFHL										
SSLH										
SSLL										
SFLH										
SFLL										
FSLH										
FSLL										

- **Include the output results after simulation, as shown in Figure 17, for reference and comparison.**
- *Explain the results (if applicable).*

EXPERIMENT 4

4.1. Objective: Understand the fanout of inverter.

4.2. Requirements: Simulate tran analysis with ADE-Explorer environment.

4.3. Instruction:

- Assemble the circuit as shown in **Figure 18**, setting parameters for power supplies (using *vpulse* source), with $L_n = L_p = 45nm$, $vdd = 1.0V$, W_N , W_P in **Experiment 1**.

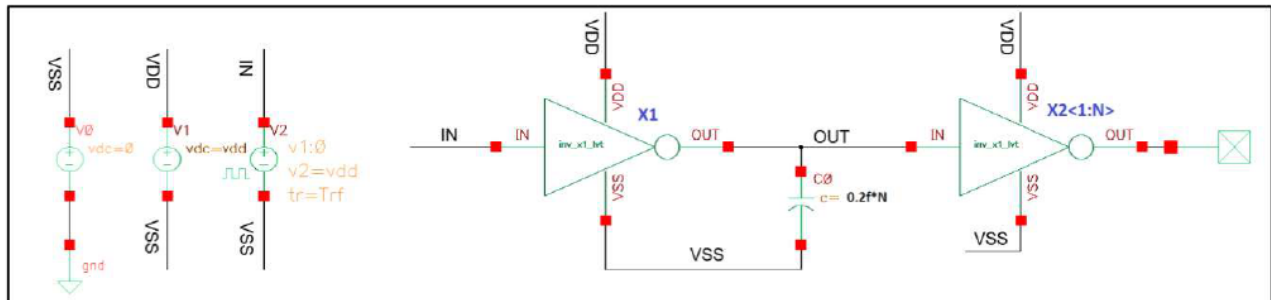


Figure 18. Testbench tran analysis of INV with fanout N

- Set the variable for *vpulse* source parameter (voltage 1, voltage 2, period, rise time, fall time, pulse width) in **Figure 13**.

4.4. Analyze and report:

➤ **4.4.1. Testcase:**

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT, FF, SS, SF, FS
Temp	-40°C, 25°C, 125°C
V_{DD}	0.9V, 1 V, 1.1V
Ln	45nm
Wn	 (Value in Experiment 1)
Lp	45nm
Wp	 (Value in Experiment 1)
Frequency	5GHz
Slew time	10%*T _{cycle}
Maximum fanout N	

- Setup testbench with 9 PVT corners in **Table 1** with ADE-Explorer.
- **Please provide the schematic and testbench here.**
- **Determine the maximum fanout N for which the inverter still operates correctly**, that means output responses still have good signal integrity (Trise, Tfall, Duty cycle and Delay).

➤ 4.4.2. Waveform

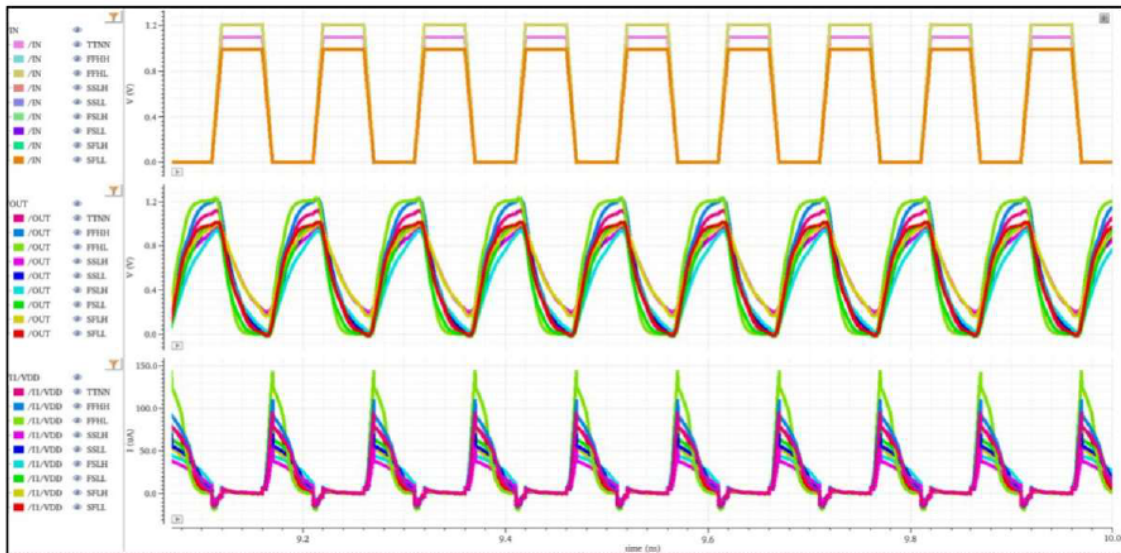


Figure 19. Waveform tran analysis of INV with fanout **N=...**

➤ 4.4.3. Observe from the waveform

- **Complete the table with the results from Calculator:**

Corner	Trise_avg (ps)	Tfall_avg (ps)	Trise (ps)	Tfall (ps)	DT_H (%)	Td_R2F (ps)	Td_F2R (ps)	I_static _0	I_static_ 1	I_dyna mic
TTNN										
FFHH										
FFHL										
SSLH										
SSLL										
SFLH										
SFLL										
FSLH										
FSLL										

- Include the output results after simulation, as shown in Figure 17, for reference and comparison.

➤ 4.4.4. *Explanation (if applicable)*

- (Students can consider additional corners to more clearly explain the variation of the inverter trip-point under PVT changes)

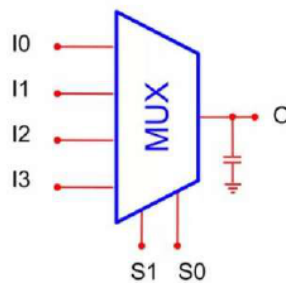
EXPERIMENT 5

5.1. Objective: Design and simulate a **4-to-1 multiplexer (MUX)** capable of operating at **5GHz** with an output load capacitance of **3fF**, using CMOS technology.

5.2. Requirements: Simulate tran analysis with ADE-Explorer environment.

5.3. Instruction:

- Design a transistor-level **4-to-1 multiplexer** using CMOS logic suitable for high-frequency operation with $L_n = L_p = 45nm$, $C_{load} = 3fF$, $v_{dd} = 1.0V$ at **9 PVT in Experiment 2**.
- Students may select any suitable **4-to-1 MUX topology** capable of operating at **5GHz** while driving a **3fF load**, such as (but not limited to):
 - Transmission-gate-based MUX
 - Inverter-Z based MUX
 - Static CMOS logic (NAND/NOR-based) MUX
 - Any other topology that meets the performance requirements



S1	S0	O
0	0	I0
0	1	I3
1	0	I1
1	1	I2

- The multiplexer must operate correctly at an input data rate (I0–I3) of **2.5GHz**, and the frequency of the **select signals** must be **2.5GHz**. The input slew time is set to **10% of the signal cycle for both I0–I3 and S0–S1**. The delay between inputs I0–I3 is **25% of the clock cycle**, and **S0 and S1 are two signals used to generate a waveform that covers all four combinations: 00, 10, 01, and 11**. **Output data will be 5.0GHz**.
- Properly size all transistors to meet speed and signal integrity requirements at 5GHz.
- Verify the functionality of the MUX through transient simulation using appropriate input and select signals.
- **Check rise time, fall time, propagation delay, voltage swing, duty cycle and power with specification**
 - **Rise time, Fall time (10%-90%) < 20ps**
 - **Duty cycle in range [47%, 53%]**
 - **Power < 5mA**

- If the circuit fails to operate correctly at 5GHz, revisit the transistor sizing and logic structure, and re-simulate until the specifications are met.

5.4. Analyze and report:

➤ 5.4.1. Testcase:

Parameter	Value
Device	nmos1v_lvt & pmos1v_lvt
Process	TT, FF, SS, SF, FS
Temp	-40°C, 25°C, 125°C
V_{DD}	0.9V, 1V, 1.1V
Frequency	5GHz
Slew time input	10%*T _{cycle}
C_{load}	3fF
Slew time (Rise/Fall time)	< 20ps
Duty cycle range	[47%, 53%]
Dynamic power	< 5mW
Topology	

- Setup testbench with 9 PVT corners in **Table 1** with ADE-Explorer.
- **Please provide the schematic and testbench here.**

➤ 5.4.2. Waveform

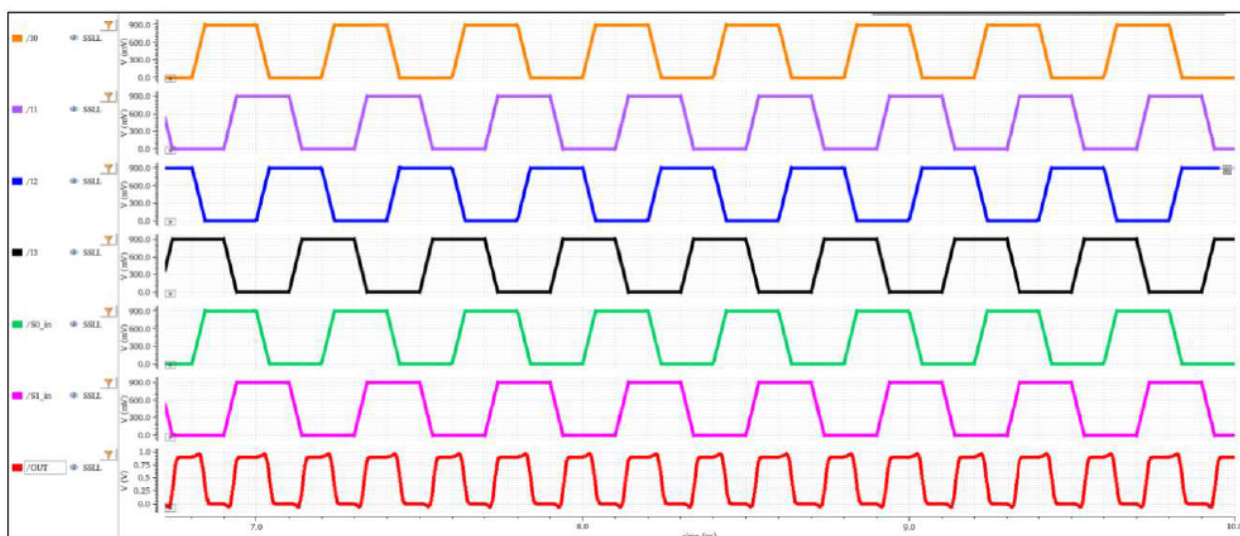


Figure 20. Waveform of MUX design at SSLL corner

➤ 5.4.3. *Observe from the waveform*

- **Complete the table with the results from Calculator:**

Corner	Trise_avg (ps)	Tfall_avg (ps)	DT_H (%)	Td_R2F (ps)	Td_F2R (ps)	I_dynamic
TTNN						
FFHH						
FFHL						
SSLH						
SSLL						
SFLH						
SFLL						
FSLH						
FSLL						

- **Include the output results after simulation for reference and comparison.**

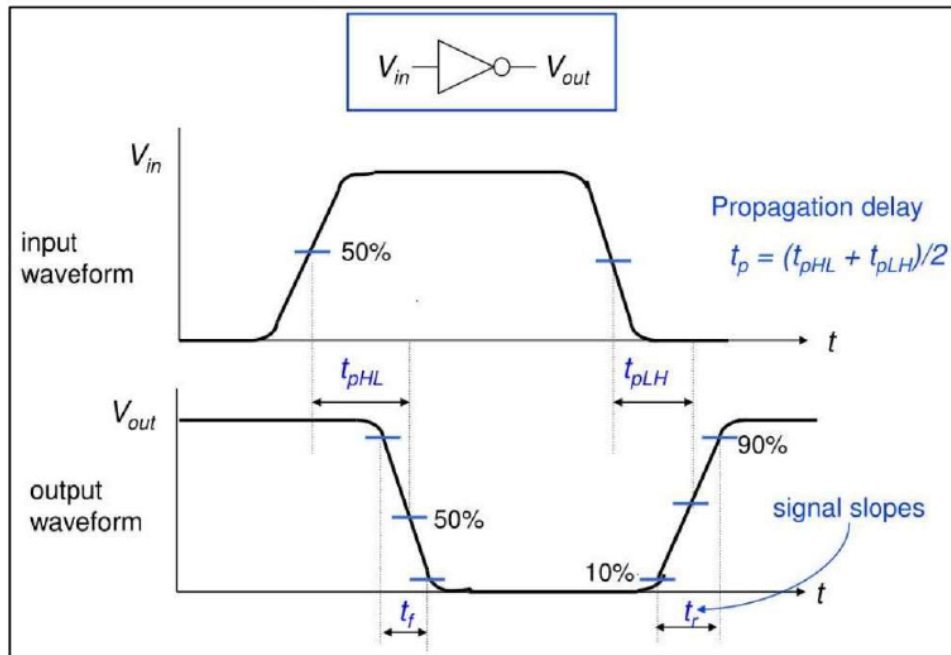
Parameter					TTNN	FFHH	FFHL	SSLH	SSLL	FSLH	FSLL	SFLH	SFLL
gpd045.scs					tt	ff	ff	ss	ss	fs	fs	sf	sf
temperature					25	125	-40	125	-40	125	-40	125	-40
vdd					1	1.1	1.1	900m	900m	900m	900m	900m	900m
Output	Spec	Pass/Fail	Min	Max	TTNN	FFHH	FFHL	SSLH	SSLL	FSLH	FSLL	SFLH	SFLL
Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
t_rise	< 20p	pass	5.123 ps	16.37 ps	7.877 ps	8.333 ps	5.123 ps	16.37 ps	9.897 ps	13.46 ps	8.044 ps	12.63 ps	7.235 ps
t_fall	< 20p	pass	4.567 ps	16.56 ps	7.322 ps	7.627 ps	4.567 ps	16.56 ps	9.252 ps	11.82 ps	6.751 ps	13.42 ps	7.372 ps
duty_cycle	range 47.53	pass	49.13 %	50.5 %	49.97 %	50.27 %	50.23 %	49.88 %	49.35 %	50.44 %	50.5 %	49.28 %	49.13 %
I_avg	< 5m	pass	2.61 mA	4.154 mA	3.225 mA	4.154 mA	4.052 mA	2.759 mA	2.61 mA	2.964 mA	2.789 mA	2.931 mA	2.762 mA

Figure 21. Measurement results of MUX design

➤ 5.4.4. *Explanation (if applicable)*

APPENDIX

❖ Slew time and skew time



Rise time is the duration a signal takes to transition from a specified low value to a high value. It is commonly defined as the time required for the waveform to rise from 10% to 90% of its final value.

In contrast, **fall time** is the duration a signal takes to decrease from a specified high value to a low value, typically measured from 90% to 10% of the peak value.

Rising propagation delay is the time taken for a signal to transition from low to high through a gate or net. It is measured from the 50% point of the input waveform to the 50% point of the output waveform.

Falling propagation delay is the time taken for a signal to transition from high to low through a gate or net. It is measured from the 50% point of the input waveform to the 50% point of the output waveform.

Propagation delay represents the total time required for a signal to travel through a gate or net, indicating how long it takes for a signal to change state.

❖ Power consumption

$$P_{dynamic} = C_L V_{DD}^2 f$$

$$P_{static} = I_{static} V_{DD}$$

$$P = P_{dynamic} + P_{static}$$

The **total power dissipation** of a circuit consists of both **dynamic** and **static** components, which can be difficult to separate in simulations. Consider the **CMOS inverter** shown below. **Dynamic power** arises from **switching currents**, which are needed to charge and discharge output loads, as well as **short-circuit currents** that briefly flow between the **PMOS and NMOS transistors** during input transitions. In contrast, **static power** results from **leakage sources** within the transistors, including **subthreshold conduction** between the source and drain, as well as **reverse-bias pn-junction leakage** between the source/drain and substrate. These leakage mechanisms are illustrated in the transistor diagram below.

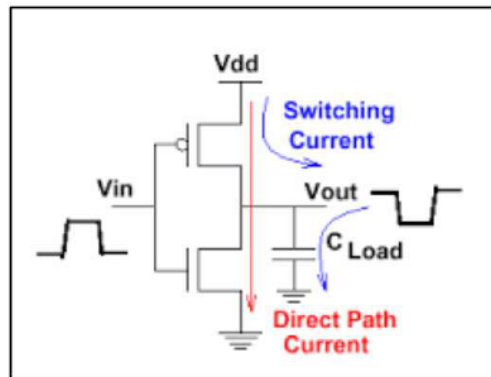


Figure 22. Dynamic power consumption.

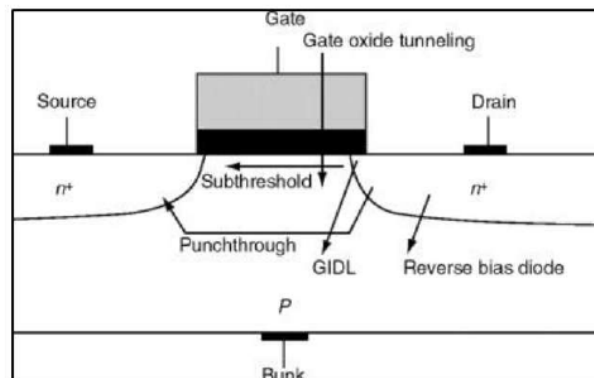


Figure 23. CMOS static leakage sources.

To measure static power dissipation, a **DC input signal** is applied to ensure no switching

occurs. In digital circuits, this means setting the input to either **high (VDD)** or **low (ground)**, which typically turns off one side of the circuit and prevents static **short-circuit current** through the transistors. For **analog circuits**, the input should be set to an appropriate **DC operating point**, usually between **ground and VDD**. Since many transistors remain turned on in their static state, analog circuits tend to **consume more static power** than digital ones.

Example: Consider a **CMOS inverter** with the following parameters: **PMOS:** 1.5 μm /0.6 μm , **NMOS:** 1.5 μm /0.6 μm , **Load capacitance:** 5pF

Using a **simple measurement method**, we can determine static power dissipation:

- **$V_{in} = 3.3\text{V} \rightarrow P_{\text{static}} = 10.94 \text{ pW}$**
- **$V_{in} = 0\text{V} \rightarrow P_{\text{static}} = 10.93 \text{ pW}$**

As expected, the power consumption is nearly the same in both cases. The slight difference is due to variations in **leakage currents** of PMOS and NMOS transistors when biased at different voltages. These differences could be more pronounced if one transistor were significantly larger than the other.

To measure total power dissipation, the circuit must **switch** between logic states, causing the output node to **charge and discharge**. In digital circuits, this requires applying a **pulse input signal**.

Example: For the same **CMOS inverter**, applying a **pulse input (pulse width = 5 μs , period = 10 μs)** results in **$P_{\text{total}} = 5.49 \text{ }\mu\text{W}$**

If the **load capacitor is removed**, the **switching current** is reduced, leaving only the current required to charge/discharge the **parasitic capacitance** at the output. As a result, the measured power will be much closer to the **static power dissipation** value.

To estimate dynamic power dissipation, subtract **static power** from **total power** to estimate the contribution of dynamic sources:

$$P_{\text{total}} = P_{\text{dynamic}} + P_{\text{static}} \rightarrow P_{\text{dynamic}}$$

➤ **Static power measurement:**

There are two methods to define the static power of an inverter: based on NMOS (when $V_{in} = 1$) or based on PMOS (when $V_{in} = 0$).

- ❖ **Dynamic power measurement:** The expression calculating dynamic power consumption:

$$P_{total} = average(u(t) \times i(t))$$

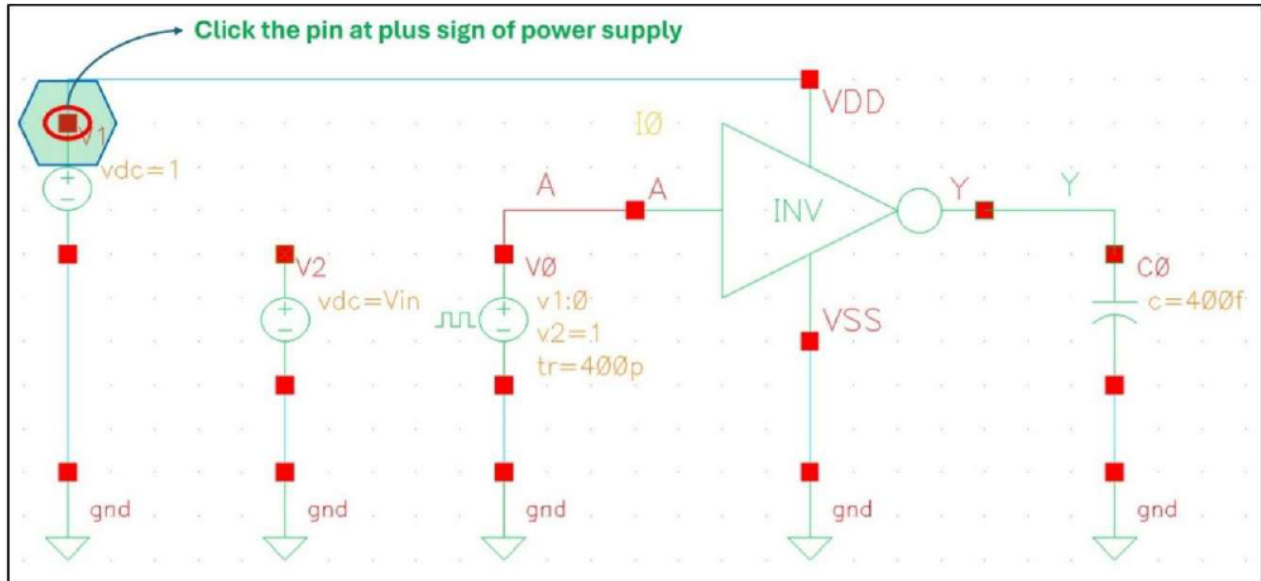


Figure 24. Choose terminal to define transient current.